



# Data Architecture

## TASCO Insurance × VETC Growth Platform — Architecture

iorta TechNXT

Version 1.0 · 07 October 2026

# Contents

1. Storage pattern
2. Conceptual model
3. Logical ERD
4. Data classification
5. Data dictionary
6. Master data management (golden record)
7. Data quality
8. Metadata, lineage and reconciliation
9. Retention, archival, erasure and backup
10. Migration strategy
11. Analytics and BI feed

| Document          | Version | Date            | Owner                      | Status                |
|-------------------|---------|-----------------|----------------------------|-----------------------|
| Data Architecture | 1.0     | 07 October 2026 | iorta TechNXT Architecture | Issued for submission |

Sources of truth:

- the collection registry `src/adapters/persistence/schema.js`;
- the DDL `db/migrations/001_init.sql` and `db/migrations/002_integrity_hardening.sql`;
- the codec `src/adapters/persistence/codec.js`;
- MDM logic `src/domain/enrichment.js`;
- rule sets `config/rules/enrichment.json`, `config/rules/service_levels.json` and `config/rules/retention.json`.

Storage pattern: [ADR-004](#). Encryption: [ADR-006](#).

## 1. Storage pattern

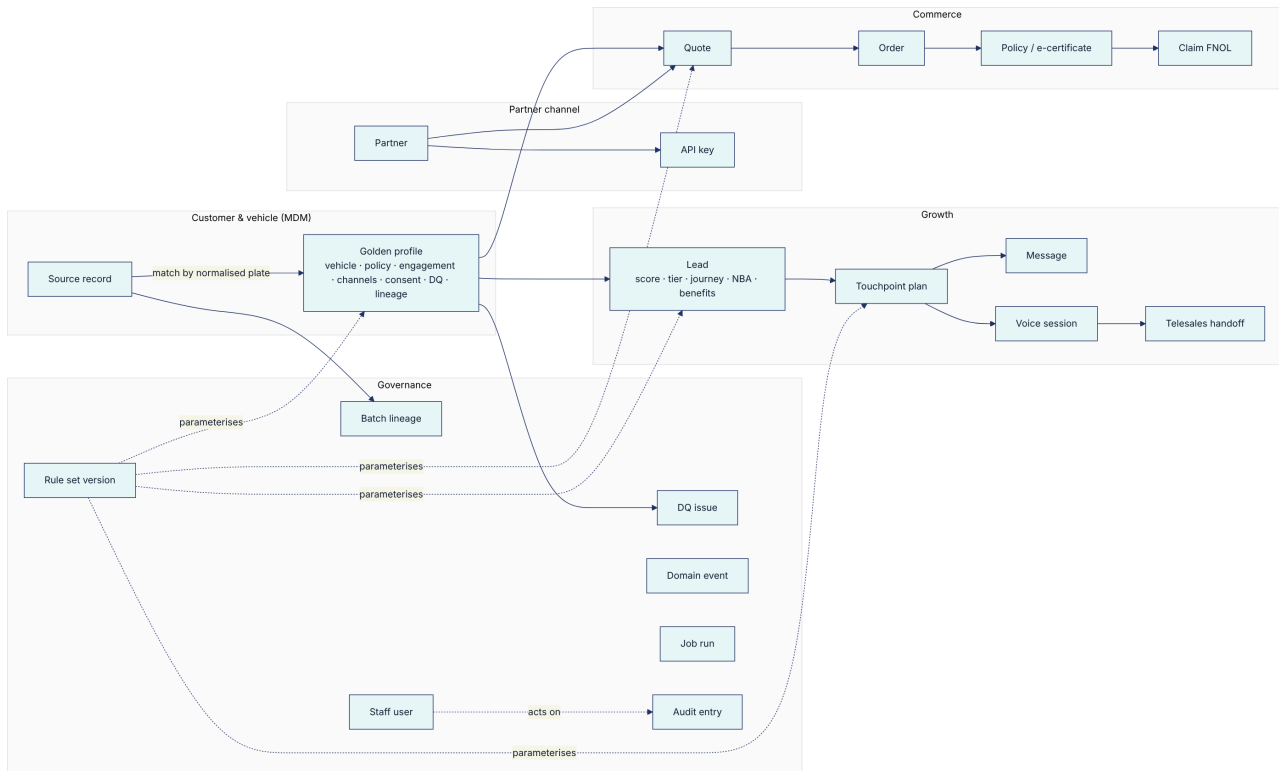
Each of the 19 collections is a PostgreSQL table with:

```
id text PRIMARY KEY · version integer (optimistic lock) · data jsonb (the document; PII fields encrypted)
· <indexed columns extracted from the document> · <blind-index columns> · created_at · updated_at
```

`audit_log` is a separate append-only, hash-chained table (UPDATE, DELETE and TRUNCATE are blocked by triggers), and `schema_migrations` records applied migrations. Only the indexed columns (plus `id`, `created_at`, `updated_at`) can be filtered or sorted (`query.js#assertColumns`).

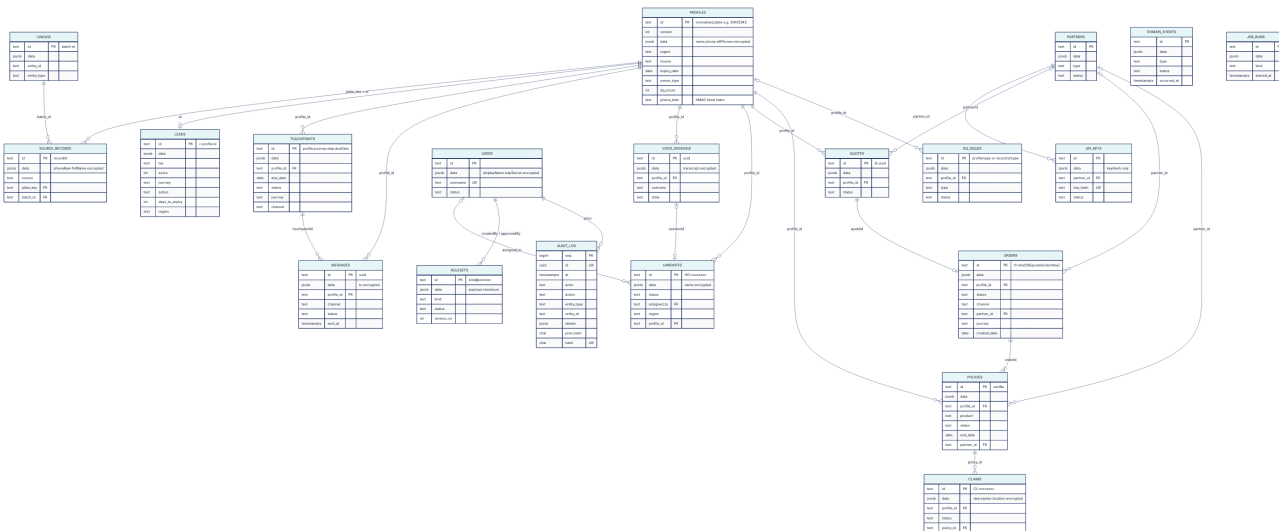
## 2. Conceptual model

The conceptual model groups the collections into five subject areas.



### 3. Logical ERD

Relationships are logical. They are enforced in the services and checked by reconciliation; there are no database foreign keys (ADR-004).



### 4. Data classification

| Class                        | Examples                                                                                    | Controls                                                                                                 |
|------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Restricted — personal</b> | Customer name and phone; call transcript; claim description and location; staff TOTP secret | AES-256-GCM field encryption; masked unless <code>profile:read_pii</code> ; redacted in logs; DSAR scope |

| Class                                 | Examples                                                                                                                       | Controls                                                                                                                                                                               |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Confidential — personal-linked</b> | Licence plate (match key), policy expiry and insurer, engagement metrics, wallet balance, consent flags, IP addresses in audit | Clear in DB (needed for matching and filtering); access by RBAC/ABAC; pseudonymised in analytics ( <b>plate status: confirm with TASCO legal / DPO</b> , see <a href="#">ADR-006</a> ) |
| <b>Confidential — business</b>        | Quotes, orders, commission, rule sets (tariffs, scoring), partner data                                                         | RBAC; audit                                                                                                                                                                            |
| <b>Internal</b>                       | Metrics, job runs, DQ issue counts                                                                                             | Staff access                                                                                                                                                                           |
| <b>Public</b>                         | Certificate validity with a masked plate (/api/public/certificates/:certNo)                                                    | Rate-limited; no PII                                                                                                                                                                   |

## 5. Data dictionary

Legend: **Idx** = extracted indexed column (filterable); **PII** = personal data; **Enc** = encrypted at rest by the codec.

"Doc" fields live only in `data.jsonb`. "Source" says which code writes the field.

## 5.1 profiles — golden record (1 per vehicle)

| Field                                                                                               | Column / type    | Idx      | PII           | Enc     | Description                                                                                   | Source                                       |
|-----------------------------------------------------------------------------------------------------|------------------|----------|---------------|---------|-----------------------------------------------------------------------------------------------|----------------------------------------------|
| id                                                                                                  | id text          | PK       | Linked        | —       | Normalised plate key, e.g. 30A12345 (normalizePlate)                                          | enrichment.buildProfiles                     |
| plate                                                                                               | doc              | —        | Linked        | —       | Display plate 30A-123.45                                                                      | same                                         |
| province                                                                                            | region text      | •        | —             | —       | Province from plate prefix (PROVINCES)                                                        | same                                         |
| name                                                                                                | doc              | —        | •             | •       | Survivor name (highest source trust)                                                          | same                                         |
| phone                                                                                               | doc + phone_bidx | • (bidx) | •             | •       | Survivor valid VN mobile 0XXXXXXXX; blind index for equality lookup                           | same                                         |
| altPhones[]                                                                                         | doc              | —        | •             | •       | Other valid phones (conflict → DQ conflicting_phone)                                          | same                                         |
| sources[], recordIds[]                                                                              | doc              | —        | —             | —       | Contributing sources and source record ids                                                    | same                                         |
| ownerType                                                                                           | owner_type text  | •        | —             | —       | individual / company                                                                          | VETC account                                 |
| tagActivatedAt                                                                                      | doc              | —        | —             | —       | ETC tag activation date (new-vehicle signal)                                                  | VETC account                                 |
| vehicle.{tollClass, seats, usage, category, categoryConfidence, categoryBasis, firstRegisteredYear} | doc              | —        | —             | —       | Vehicle facts; category from enrichment.categoryTable                                         | MDM                                          |
| policy.expiryDate                                                                                   | expiry_date date | •        | Linked        | —       | Best expiry estimate                                                                          | MDM / customer / steward / bot / TASCO issue |
| policy.{expiryMethod, expiryConfidence, expiryCandidates[], certNo, verified}                       | doc              | —        | —             | —       | Evidence and confidence                                                                       | same                                         |
| policy.insurer                                                                                      | insurer text     | •        | —             | —       | Current insurer (TASCO, other, OTHER, null)                                                   | same                                         |
| policy.competitorNote                                                                               | doc              | —        | • (free text) | — (gap) | Customer's own words about the competitor policy (voice bot); cleared on erasure              | voiceService.finalize                        |
| engagement.*                                                                                        | doc              | —        | Linked        | —       | App sessions, toll trips, long trips, wallet balance, auto top-up, prior purchase, complaints | VETC account                                 |
| channels.{app_push, zalo_zns, sms, voice_bot, telesales}                                            | doc              | —        | —             | —       | Reachability                                                                                  | MDM                                          |

| Field                          | Column / type | Idx | PII | Enc | Description                                         | Source                    |
|--------------------------------|---------------|-----|-----|-----|-----------------------------------------------------|---------------------------|
| consent.{marketing, call, dnc} | doc           | —   | •   | —   | Consent state                                       | VETC + platform overrides |
| consentOverrides               | doc           | —   | •   | —   | Platform-captured consent that survives rebuilds    | customer / bot opt-out    |
| dataQuality.score              | dq_score int  | •   | —   | —   | 0–100 (§7)                                          | MDM                       |
| dataQuality.missing[]          | doc           | —   | —   | —   | Open DQ facets                                      | MDM                       |
| lineage[]                      | doc           | —   | —   | —   | {field, source, confidence, rule?, at?}             | MDM + updates             |
| partnerId                      | doc           | —   | —   | —   | Partner that supplied a record                      | MDM                       |
| anonymised, anonymisedAt       | doc           | —   | —   | —   | Erasure markers (rebuild skips anonymised profiles) | customerService.erase     |

## 5.2 source\_records — landed raw records

| Field                                                                                                                                                                                                             | Column          | Idx | PII              | Enc | Description                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| id = recordId                                                                                                                                                                                                     | PK              | •   | —                | —   | Source - unique record id                                                                                                                 |
| source                                                                                                                                                                                                            | source          | •   | —                | —   | vetc_account, partner_<type>, telesales_csv, ...                                                                                          |
| plateRaw / plateKey                                                                                                                                                                                               | doc / plate_key | •   | Linked           | —   | Raw and normalised plate (null if invalid)                                                                                                |
| phoneRaw, fullName                                                                                                                                                                                                | doc             | —   | •                | •   | Raw contact data                                                                                                                          |
| batchId                                                                                                                                                                                                           | batch_id        | •   | —                | —   | Ingestion batch (B-<date>-<rand>)                                                                                                         |
| tollClass, seatsDeclared, usageDeclared, ownerType, tagActivatedAt, firstRegisteredYear, policy{insurer, certNo, expiryDate, verified}, lastInspectionDate, declaredExpiry, partnerId                             | doc             | —   | Linked           | —   | Source attributes                                                                                                                         |
| VETC account attributes (appUser, appSessions30d, tollTrips30d, longTripsKm90d, walletBalance, autoTopUp, pushEnabled, zaloLinked, marketingConsent, callConsent, dnc, complaints12m, priorVetcInsurancePurchase) | doc             | —   | Linked / consent | —   | Engagement, channel and consent. Accepted by POST /api/data/ingest (key allow-list; value types not yet checked) and by the bulk/CDC path |
| _hidden                                                                                                                                                                                                           | doc             | —   | —                | —   | <b>Synthetic data only</b> (ground truth for model evaluation). Must never exist in production feeds. Stripped from DSAR export.          |
| ingestedAt, anonymised                                                                                                                                                                                            | doc             | —   | —                | —   | Landing time; erasure marker                                                                                                              |

## 5.3 leads

| Field                                                   | Column         | Idx | Description                                                   |
|---------------------------------------------------------|----------------|-----|---------------------------------------------------------------|
| id = profileId                                          | PK             | •   | One lead per profile                                          |
| tier                                                    | tier           | •   | hot ≥ 70, warm ≥ 45, else nurture (scoring.tiers)             |
| score                                                   | score          | •   | 0–100, damped by expiry confidence; 0 when DNC                |
| journey, objective                                      | journey        | •   | First matching journey (journeys.priority)                    |
| nextBestAction.<br>{action, label, reason, ruleId, ...} | action         | •   | NBA decision-table output                                     |
| daysToExpiry                                            | days_to_expiry | •   | Negative = lapsed                                             |
| region                                                  | region         | •   | Province                                                      |
| reasons[]                                               | doc            | —   | Per-factor {factor, label, points, max, why} (explainability) |
| benefits[], premium, plate, evaluatedAt                 | doc            | —   | Top benefits, TNDs premium incl. VAT, business date           |

## 5.4 Engagement collections

| Collection     | Key fields (Idx in bold)                                                                                                                                                                                                                                                                     | PII / Enc                                                                                                                                     | Notes                                                                                                                                                                                                 |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| touchpoints    | id (profile:journey:step:dueDate), <b>profile_id</b> , <b>due_date</b> , <b>status</b> , <b>journey</b> , <b>channel</b> , step, channels[], marketing, template, result, executedAt                                                                                                         | —                                                                                                                                             | Deterministic id gives idempotent planning. Status scheduled → done/skipped/cancelled. A touchpoint blocked only by the contact window stays scheduled (deferred), with lastAttemptAt and lastResult. |
| messages       | id uuid, <b>profile_id</b> , <b>channel</b> , <b>status</b> , <b>sent_at</b> , templateKey, journey, step, touchpointId, marketing, text, blockReason, providerMessageId, error, sessionId                                                                                                   | to • Enc. text contains the plate and a renewal link (Linked, clear).                                                                         | Status sent/failed/blocked. Contact history for caps.                                                                                                                                                 |
| voice_sessions | id uuid, <b>profile_id</b> (customerId), <b>outcome</b> , <b>state</b> , mode, verified, plateAttempts, clarifications, signals{priceAsked, trustConcern, expiryConfirmed, competitorInfo, expiryStatement}, ctx{plateKey, plateMasked, expiry, ...}, persona, handoffId, startedAt, endedAt | transcript • Enc. signals.competitorInfo and signals.expiryStatement are free-text customer speech stored in clear (gap); cleared on erasure. | Retention 180 d                                                                                                                                                                                       |
| handoffs       | id, <b>status</b> , <b>assigned_to</b> , <b>region</b> , <b>profile_id</b> , sessionId, plate, plateVerifiedByCustomer, phoneMasked, journey, expiryDate, daysToExpiry, premium, score, outcome, priceAsked, trustConcern, talkingPoints[], notes[], createdAt                               | name • Enc; phone masked                                                                                                                      | Minimal data for telesales (handoffSummary)                                                                                                                                                           |

## 5.5 Commerce collections

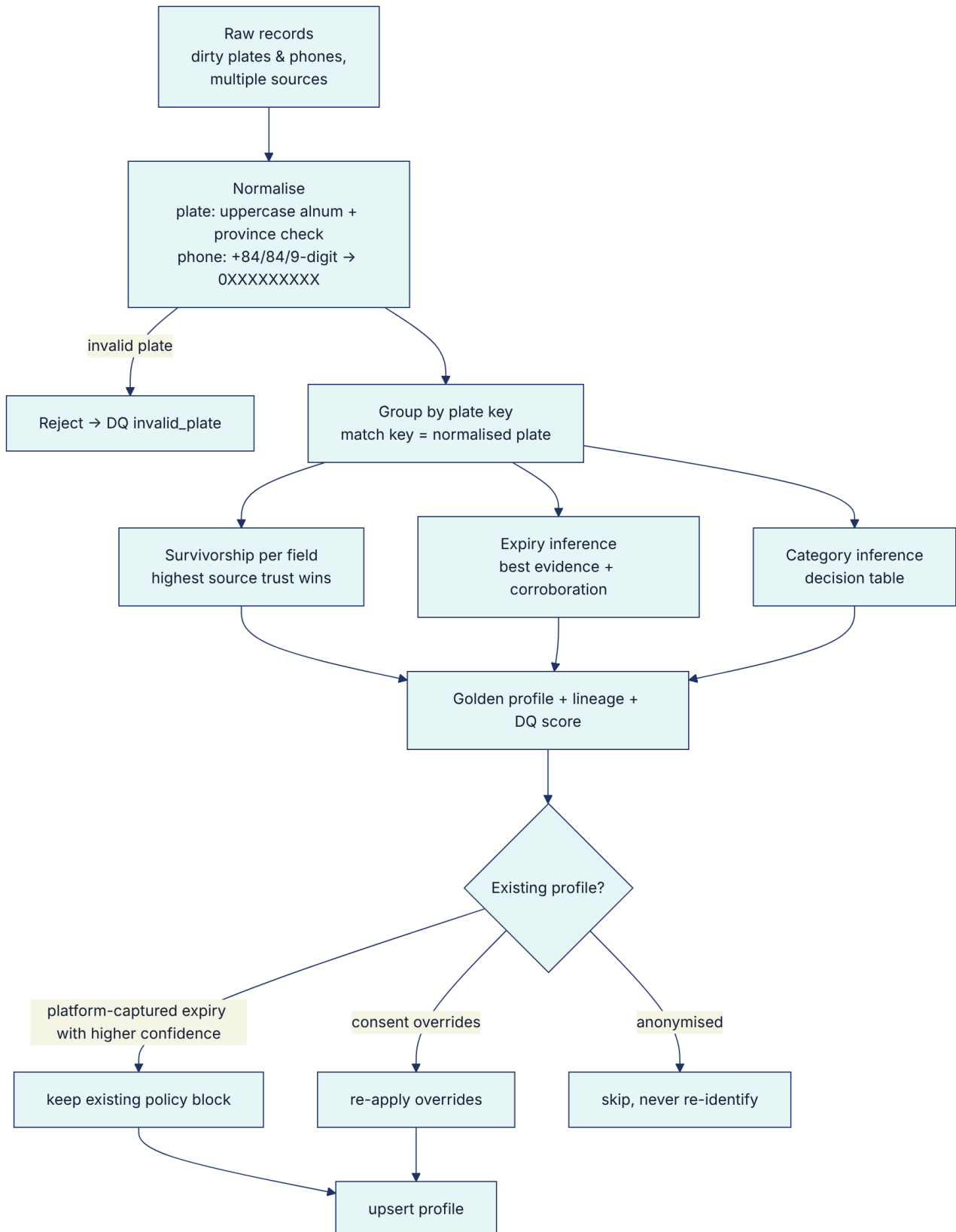
| Collection | Key fields (Idx in bold)                                                                                                                                                                                                                                                                                                    | PII / Enc      | Notes                                                                                                                                                            |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| quotes     | id Q-uuid, <b>profile_id</b> , <b>status</b> , plate, channel, partnerId, journey, lines[{product, startDate, endDate, termDays, premiumNet, vat, total, priceRegulated, breakdown}], total, benefits, bundle, createdBy, expiresAt, inspection{passed, evidence, by, at}, sentToCustomerAt, sentBy, payingOrderId, orderId | Linked (plate) | TTL quoteTtlHours (24 h, service_levels). Status open → paying → converted; paying returns to open if payment or issuance fails.                                 |
| orders     | id 0-..., <b>profile_id</b> , <b>status</b> , <b>channel</b> , <b>partner_id</b> , <b>journey</b> , <b>created_date</b> , quoteId, amount, paymentRef, policies[], commission[{product, rate, amount, ruleId, capped}], refund{status, refundId}, cancelFailures[], createdBy, completedAt, error                           | —              | pending_payment → completed, or payment_failed, issuance_failed_refunded or compensation_failed. Partner orders have paymentRef PARTNER-... (partner-collected). |
| policies   | id = certNo, <b>profile_id</b> , <b>product</b> , <b>status</b> , <b>end_date</b> , <b>partner_id</b> , policyNo, plate, startDate, premiumNet, vat, total, channel, orderId, certificateUrl, issuedAt, insurer                                                                                                             | Linked (plate) | Financial and legal record (10-year archive)                                                                                                                     |

| Collection | Key fields (Idx in bold)                                                                                             | PII / Enc                   | Notes                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------|
| claims     | id CL-..., <b>profile_id</b> , <b>status</b> , <b>policy_id</b> , product, incidentDate, photos, history[], slaDueAt | description, location • Enc | FNOL front door; slaDueAt from claimAckSlaHours (4 h). Adjudication happens in TASCO core. |

## 5.6 Partner, identity and governance collections

| Collection    | Key fields (Idx in bold)                                                                                                                                                                                                     | PII / Enc                                                                     | Notes                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| partners      | id, <b>type</b> , <b>status</b> , name, region, createdAt                                                                                                                                                                    | —                                                                             | Types: bank, showroom, agent, fleet, inspection_center                                                                                   |
| api_keys      | id, <b>partner_id</b> , <b>key_hash (unique)</b> , <b>status</b> , prefix, scopes[], createdBy, revokedAt                                                                                                                    | — (SHA-256 only)                                                              | Plaintext key returned once                                                                                                              |
| users         | id, <b>username (unique)</b> , <b>status</b> , roles[], region, passwordHash (scrypt), failedLogins, lockedUntil, mfaEnrolled, lastTotpStep, tokensValidAfter, mustChangePassword, lastLoginAt, createdBy, passwordChangedAt | displayName, totpSecret • Enc                                                 | Staff only. lastTotpStep blocks TOTP replay; tokensValidAfter revokes earlier tokens; mustChangePassword is set for admin-created users. |
| rulesets      | id kind@N, <b>kind</b> , <b>status</b> , <b>version_no</b> , payload, checksum, description, createdBy, submittedAt, approvedBy, approvalComment, activatedAt, retiredAt, supersededBy, rejectedBy, rejectionComment         | —                                                                             | Maker-checker history. Unique partial index: one active version per kind (migration 002).                                                |
| domain_events | id, <b>type</b> , <b>status</b> , <b>occurred_at</b> , payload, actor, correlationId, attempts, handled[], lastError, processedAt                                                                                            | Payload carries profile ids (plates)                                          | Outbox; index on (status, occurred_at) (migration 002)                                                                                   |
| dq_issues     | id, <b>profile_id</b> , <b>type</b> , <b>status</b> , recordId, source, batchId, detectedAt, via, resolution, resolvedBy, resolvedAt                                                                                         | —                                                                             | \$7                                                                                                                                      |
| lineage       | id = batchId, <b>entity_id</b> , <b>entity_type</b> , source, records, rejected, at                                                                                                                                          | —                                                                             | Batch lineage                                                                                                                            |
| job_runs      | id, <b>kind</b> , <b>started_at</b> , actor, status, result, error, finishedAt                                                                                                                                               | —                                                                             | Job history                                                                                                                              |
| audit_log     | seq, id, at, actor, action, entity_type, entity_id, details, prev_hash, hash                                                                                                                                                 | details hold ids, reasons and IP addresses (personal data of staff/customers) | Append-only, hash-chained                                                                                                                |

## 6. Master data management (golden record)



**Match key.** `normalizePlate()` ( `src/domain/identity.js` ) produces `^\d{2}[A-Z]{1,2}\d{4,5}$` with a valid province code. It handles forms such as `30a-123.45`, `30A 12345`, `30A.123.45` and the legacy 4-digit form. Exact match on the key only; there is no fuzzy matching. That is deliberate, because a plate is a legal identifier. Wrong-person and plate-mismatch outcomes from calls feed DQ.

**Survivorship** ( `enrichment.sourceTrust` ):

| Source                        | Trust |
|-------------------------------|-------|
| tasco_core                    | 1.00  |
| vetc_app_purchase             | 0.95  |
| vetc_account                  | 0.85  |
| customer_declared             | 0.75  |
| partner_inspection_center     | 0.65  |
| partner_bank, showroom, fleet | 0.60  |
| partner_agent                 | 0.50  |
| telesales_csv, default        | 0.40  |

| Attribute                                     | Rule                                                                                                                                                        |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phone                                         | Valid normalised phones sorted by trust. The first is the survivor; the rest go to <code>altPhones</code> . More than one distinct phone raises a DQ issue. |
| name, seats, usage, tollClass                 | Highest-trust non-empty value                                                                                                                               |
| ownerType, tag, engagement, channels, consent | From the <code>vetc_account</code> record (system of record for the account)                                                                                |
| vehicle.category                              | <code>categoryTable</code> (first hit) over seats, toll class and commercial flag, with confidence and basis                                                |
| policy.expiryDate                             | Evidence ranking (below)                                                                                                                                    |
| policy.insurer                                | First record carrying an insurer, else the insurer from the best expiry evidence                                                                            |

**Expiry evidence** ( `enrichment.expiryEvidence` , `inferExpiry` ):

| Method                | Confidence                                                           |
|-----------------------|----------------------------------------------------------------------|
| verified_certificate  | 1.0                                                                  |
| customer_declared     | 0.75                                                                 |
| partner_policy_record | $0.7 \times \text{trust} / 0.6$ (capped at 0.7)                      |
| inspection_cycle      | 0.5 (last inspection + 365 d, rolled forward to $\geq$ today - 60 d) |
| tag_anniversary       | 0.25                                                                 |

- **Corroboration:** each different method agreeing within 21 days adds 0.15, up to 0.95.
- **Usable** means confidence  $\geq 0.5$ .

Facts captured on the platform are applied by `applyDeclaredExpiry`. Their confidences are governed in the rule kind `service_levels` (maker-checker), except TASCO issuance:

| Platform source | Confidence | Where defined                                           |
|-----------------|------------|---------------------------------------------------------|
| customer (app)  | 0.75       | <code>service_levels.customerDeclaredConfidence</code>  |
| data steward    | 0.9        | <code>service_levels.stewardCorrectionConfidence</code> |

| Platform source               | Confidence | Where defined                                             |
|-------------------------------|------------|-----------------------------------------------------------|
| voice bot (renewed elsewhere) | 0.6        | <code>service_levels.botRenewedElsewhereConfidence</code> |
| TASCO issuance                | 1.0        | <code>salesService</code> (verified certificate)          |

**Rebuild protection.** On rebuild, an existing policy block is kept when its method is `customer_declared`, `voice_bot`, `tasco_issued` or `data_steward` and it has higher confidence ( `ingestionService.rebuild` ). A steward correction therefore survives the next rebuild from source data.

## 7. Data quality

**Score** (per profile, 0–100):

```
completeness = 1 - (#missing of {phone, name, reliable_expiry, vehicle_category, current_insurer}) / 5
score = round(100 × (0.5·completeness + 0.35·expiryConfidence + 0.15·categoryConfidence)) //
weights: enrichment.dataQuality
```

| DQ dimension              | Measure                                                       | Issue type(s)                                                         | Raised by                            |
|---------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------|
| Validity                  | Plate and phone format, province                              | <code>invalid_plate</code> (record-level)                             | <code>ingestionService.ingest</code> |
| Completeness              | Mandatory facets present                                      | <code>phone</code> , <code>name</code> , <code>current_insurer</code> | <code>buildProfiles</code>           |
| Accuracy / reliability    | Expiry confidence $\geq 0.5$ ; category confidence $\geq 0.6$ | <code>reliable_expiry</code> , <code>vehicle_category</code>          | <code>buildProfiles</code>           |
| Consistency               | One phone per vehicle                                         | <code>conflicting_phone</code>                                        | <code>buildProfiles</code>           |
| Accuracy (field-verified) | Customer contradicts the data                                 | <code>wrong_person</code> , <code>plate_mismatch</code>               | <code>voiceService.finalize</code>   |
| Uniqueness                | Duplicates merged per plate                                   | <code>stats.duplicatesMerged</code>                                   | <code>buildProfiles</code>           |
| Timeliness                | Freshness of the source feed                                  | (planned: <code>stale_source</code> )                                 | Feed monitoring                      |

### Workflow:

- Issues are listed at `GET /api/dq/issues` ( `dq:read` ).
- A data steward resolves them via `POST /api/dq/issues/:id/resolve` ( `dq:resolve` ), or corrects the expiry with `PATCH /api/customers/:id/expiry` (evidence required, audited).
- Customers fix their own data ( `POST /api/customer/expiry` ), and calls fix it too.
- A resolved issue reopens if the next rebuild detects it again.

KPIs ( `/api/dashboard/overview` ): open issues by type, and profiles with usable data. `profilesWithUsableData` = profiles minus open `reliable_expiry` issues, which matches the MDM definition (expiry confidence  $\geq 0.5$ ).

## 8. Metadata, lineage and reconciliation

### Technical metadata.

- `schema.js` (collections, indexed columns, PII and blind-index fields) is machine-readable.
- Target:** export it to the data catalogue (e.g. OpenMetadata/DataHub) with classification tags from §4.
- OpenAPI ( `/api/openapi.json` ) documents interface metadata.

## Business metadata.

- Each rule set version carries `kind`, `description`, `checksum`, `createdBy`, `approvedBy`, `activatedAt`.
- The descriptions in `config/rules/*.json` act as a business glossary seed.

## Lineage, at four levels:

| Level            | Where                                                                                                                                    | API                                                         |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Batch            | <code>lineage collection {batchId, source, records, rejected, at}; source_records.batchId</code>                                         | GET <code>/api/customers/:id/lineage (recentBatches)</code> |
| Record → profile | <code>profiles.recordIds[], sources[]</code>                                                                                             | same                                                        |
| Field            | <code>profiles.lineage[] {field, source, confidence, rule, at}</code>                                                                    | same                                                        |
| Decision         | <code>leads.reasons[] (factor, points, why), nextBestAction.ruleId, quote.lines[].breakdown[].rateRule, order.commission[].ruleId</code> | GET <code>/api/customers/:id, /api/leads</code>             |

**Gap:** leads and quotes do not record **which rule-set versions** (`kind@N` or `checksum`) produced them. Add a `rulesSnapshot` (`rulesService.snapshot()` already returns `{kind, version, checksum}`) to leads, quotes and orders, so any decision can be reproduced for regulators.

## Reconciliation:

| Check                                                                                                                                                                                                                                       | Implemented |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Ingestion: records = landed + rejected; profiles = distinct valid plates (returned per batch)                                                                                                                                               | •           |
| Orders ( <code>opsService.reconcile</code> ): completed ⇒ payment ref present and each policy exists; <code>pending_payment &gt; 1 h</code> , <code>payment_failed</code> (confirm no capture) and <code>compensation_failed</code> flagged | •           |
| Wallet settlement ↔ orders; TASCO core policy register ↔ policies; partner commission ↔ finance                                                                                                                                             | Planned     |
| Event backlog / dead letters ( <code>/api/ops/status</code> )                                                                                                                                                                               | • (manual)  |
| Audit chain integrity ( <code>/api/audit/verify</code> )                                                                                                                                                                                    | •           |

## 9. Retention, archival, erasure and backup

**Retention policy** (`config/rules/retention.json`, rule kind `retention`, maker-checker; **periods to be confirmed with TASCO legal**):

| Entity                  | Retain        | Action                                          | Implemented by<br><code>opsService.applyRetention</code>        |
|-------------------------|---------------|-------------------------------------------------|-----------------------------------------------------------------|
| source_records          | 365 d         | delete                                          | • <code>deleteWhere(created_at &lt; cutoff)</code> — see note 1 |
| profiles                | 1,825 d (5 y) | anonymise when no active policy and no activity | Planned (archival pipeline)                                     |
| voice_sessions          | 180 d         | delete (recordings + transcripts)               | •                                                               |
| messages                | 365 d         | archive                                         | Planned                                                         |
| orders                  | 3,650 d       | archive (financial records)                     | Planned                                                         |
| certificates (policies) | 3,650 d       | archive                                         | Planned                                                         |

| Entity    | Retain  | Action                       | Implemented by |
|-----------|---------|------------------------------|----------------|
| audit_log | 3,650 d | archive, never deleted early | Planned (WORM) |

#### Notes:

1. `created_at` is preserved on upsert, so source-record age counts from **first landing**, not from the last refresh. A record refreshed daily by CDC is still deleted after 365 days, after which the profile rebuild loses that source. Use `updated_at`, or keep the latest version per source.
2. The policy description says "legal hold overrides deletion", but **no legal-hold flag exists in code**. Add `legalHold` on profiles, claims and orders, and skip held entities.
3. `domain_events`, `touchpoints`, `dq_issues` and `job_runs` have no retention rule. Add 30 d ( `done` events), 90 d (executed touchpoints), 365 d (resolved DQ) and 365 d (job runs).

**Erasure** ( `customerService.erase`, `POST /api/dsar/:id/erase`, `dsar:manage` ):

- Refused while an active policy exists (legal retention).
- Otherwise it:
  - nulls the profile's name, phone and `competitorNote`, and sets `anonymised`, DNC and no consent;
  - deletes the lead;
  - nulls PII in up to 100 source records;
  - erases message `to` and text;
  - empties voice transcripts and clears the free-text signals ( `competitorInfo`, `expiryStatement` );
  - clears handoff name, masked phone and notes;
  - replaces claim description and location;
  - writes an audit entry.
- Erased data subjects lose their customer sessions, and their renewal links stop working.
- **Export** ( `exportData`, `POST /api/dsar/:id/export`, `GET /api/customer/data-export` ) includes the profile, lead, policies, messages, source records, voice sessions, quotes, orders, claims and telesales tasks (handoffs), with synthetic `_hidden` fields stripped.
- **Open points:**
  - claim records follow claims retention (**confirm with TASCO legal**);
  - VETC must also be notified (suppression list), otherwise the next feed re-lands contact data in `source_records`. The profile stays anonymised because the rebuild skips anonymised profiles.

**Backup.** Managed PITR (14 d) plus monthly snapshots (12 months), encrypted with KMS, in the same jurisdiction (Deployment and Infrastructure Architecture, section 6). Field-encrypted PII stays encrypted in backups, so **data keys must be retained for as long as backups that need them**. A key may only be destroyed after the re-key job has run and the backups have aged out. Erasure in backups follows "beyond use" handling: restored backups re-apply erasures from the audit log ( `dsar.erased` entries) before use.

## 10. Migration strategy

### Schema migrations.

- Numbered SQL files in `db/migrations/NNN_description.sql` (today `001_init.sql` and `002_integrity_hardening.sql` ), applied in order and recorded in `schema_migrations`, each in its own transaction, under a PostgreSQL advisory lock so concurrent runners cannot race

```
( postgresStore.migrate ).
```

- Never edit an applied file.
- The registry ( `schema.js` ) and the DDL are kept in step by the schema test.
- Expand/contract for zero-downtime releases (Deployment and Infrastructure Architecture, section 10).
- Run them from the single migration Job in Kubernetes ( `MIGRATE_ON_START=false` on replicas) or the Railway pre-deploy command.
- **Document shape changes** inside `data` need no DDL. Readers must tolerate missing fields; prefer additive changes. For breaking changes, use a backfill job with `update()` and keyset paging.

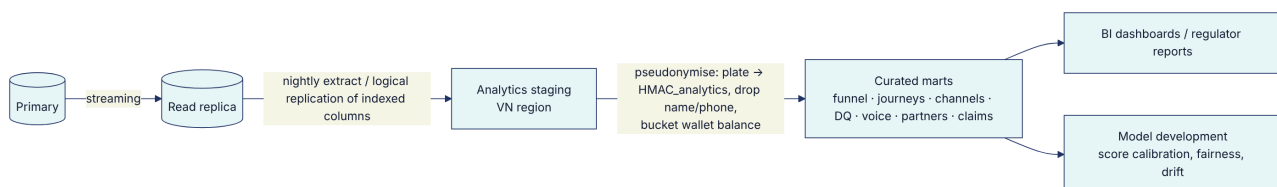
#### Legacy data migration:

| Source                                        | Approach                                                                                                                                                                                                                      | Trust / evidence                     |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| VETC account base (6 M)                       | Bulk file load → staging → parallel rebuild (Integration Architecture, section 6)                                                                                                                                             | vetc_account 0.85                    |
| TASCO core in-force and expired TNDs policies | Extract the policy register (plate, cert no, period, holder) as source <code>tasco_core</code> with <code>policy.verified = true</code> . This makes renewal journeys ( <code>insurer = TASCO</code> ) accurate from day one. | tasco_core 1.0, verified certificate |
| VETC app past insurance purchases             | Source <code>vetc_app_purchase</code>                                                                                                                                                                                         | 0.95                                 |
| Partner and telesales lists                   | Only with a documented lawful basis ( <b>confirm with TASCO legal</b> ); sources <code>partner_&lt;type&gt;</code> / <code>telesales_csv</code>                                                                               | 0.4–0.65                             |
| Consent and DNC registers                     | Loaded <b>before</b> any journey is enabled; consent overrides take precedence                                                                                                                                                | —                                    |

#### Migration steps:

1. Profile the source data.
2. Agree the mapping with data owners.
3. Run trial loads in SIT with masked data.
4. Reconcile counts and samples with the business.
5. Dress-rehearse in perf.
6. Production load in a freeze window.
7. Run the post-load DQ report.
8. Enable journeys progressively by region.

## 11. Analytics and BI feed



#### Rules:

- **No decrypted PII leaves the platform** for analytics.
- The analytics pseudonym uses its own key, never `BLIND_INDEX_KEY`.
- Join keys are stable pseudonyms.

- Small-cell suppression ( $n < 10$ ) applies in reports shared outside the data office.
- Event-level facts ( `messages` , `touchpoints` , `voice_sessions` outcomes, `orders` ) feed attribution (journey → order via `orders.journey` ).

The in-product dashboards ( `insightsService.overview/adoption/governance` ) cover operational KPIs. Heavy analysis belongs in the marts.